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Review

A review on the development of optical limiters from homogeneous to reflective 1-D photonic crystal structures

Reena Gadhwal, Ambika Devi *

Department of Applied Sciences, The NorthCap University, Sector 23-A, Gurugram 122017, Haryana, India

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ABSTRACT

The optical limiting behavior of numerous nonlinear materials has been reviewed. Explicit examination of nonlinear mechanisms such as multiphoton absorption, nonlinear absorption, nonlinear scattering and their peculiar implementation in optical devices using nonlinear materials as carbon nanotubes, fullerenes, semiconductors, graphene, and carbon black suspensions have been performed. Reflection-based optical limiters are especially emphasized for their self-protecting optical limiting over a broad transmission bandwidth. A comparative study of limiting parameters of several nonlinear materials is also included.

1. Introduction

The proliferated lasers and their implementation in a plethora of applications [1,2] have promoted the scope of novel laser-protecting optical switches and optical limiting devices. The majority of devices are assimilating low-intensity lasers like CCDs, DVDs, and thermal cameras while high-intensity laser radiations are confined to defense, research, astronomy, medical, and industries [3,4]. The explicit widening of laser weapons in military applications comes with the limitation of hazards to human eyes and sophisticated equipment. Manipulating the intensity of light in a foreseeable and pre-established manner can protect such a delicate and light-sensitive device. Though a variety of mechanisms have been invented to modulate, limit, regulate, or switch light signals, all can be categorized broadly as a dynamic and passive mechanism [5]. Subsequently, optical limiter, an optical device to limit the intensity above a fluence threshold by suppressing undesired radiation can also be classified as a dynamic optical limiter and passive optical limiter. An Iris of the human eye or a neutral density filter being such kind of dynamic limiter depends on active feedback for their processing. The dependency of components on each other for the limiter processing makes their mechanism highly complicated and sluggish at the same time with reduced efficiency. On the contrary, passive limiting relies solely on the inherent physical properties of their intelligent material, enhancing the speed and performance of the optical device. Optical limiters have been extensively probed for a variety of applications in pulse compression, pulse switching, pulse smoothing, pulse shaping, and mode-locking [6–8]. However, an optical limiter has exclusive importance in protecting light-sensitive devices and the human eye from intense laser irradiance [9–11]. For many years, several research groups have employed lots of strategies for developing potent optical limiters [12,13]. A novel optical limiter demands few specifications as low activating threshold and high damage threshold, broadband operating response for any wavelength, and should be expeditious for any pulse duration [14]. The immanent nonlinearities like reverse saturation absorption [15–17], multiphoton absorption [18], nonlinear refraction [19], nonlinear scattering [14], and their amalgamation [20–22] have been extensively investigated by many researchers for over two decades for developing optical limiters. To address the problem of overheating and hence sacrificing itself in the process of protecting a delicate optical sensor by limiting materials, Reflection based optical limiter has manifested a new class of self-protecting optical limiters. These limiters are consisting of a 1-D photonic multilayered structure for reflecting the high-intensity radiation and protecting the valuable delicate optical sensors (see Fig. 1).

the action, process or results of combining

Several reviews on the history and contemporary activities in the evolution of limiting materials and optical limiting devices have already been published [5,23–25]. Here we furnish the review on optical limiting materials and the mechanism relying on a somewhat different perspective. Inherent nonlinearities as reverse saturable absorption (RSA) in fullerenes and dye molecules, multiphoton absorption (MPA) in chalcogenide and semiconductors, nonlinear scattering (NLS) in carbon nanotubes, nanocomposites, and colloidal solutions, and many others

* Corresponding author.
E-mail address: ambikasharma2004@yahoo.co.in (A. Devi).

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fluence = a stream of particles crossing a unit area, usually expressed as the number of particles per second.

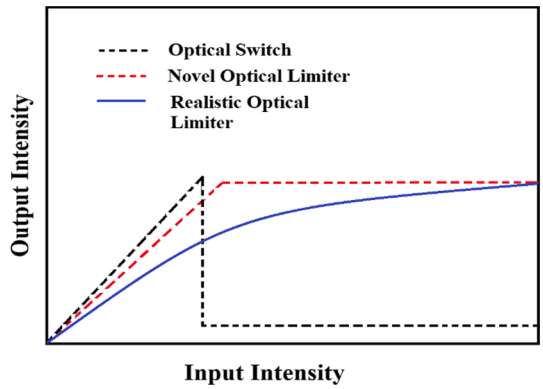

Fig. 1. The nonlinear response of an optical limiter and optical switch.

are covered here along with the limitation of their vulnerability and self-sacrifice against high irradiance. Central attention is captured by multilayer structured reflection-based limiters. Finally, a comparison of optical limiting properties of different materials and their mechanism is also included.

2. Mechanism for optical limiting

Leite et al. [26] demonstrated experimentally for the first time as early as 1967 the thermal lens effect-based optical power limiter which was induced by laser irradiance. 488 nm CW laser beam was used as incident light and nitrobenzene was used as a linear absorbing medium providing the base idea for a very popular optical limiting design. To date, with the same base design, a variety of materials (semiconductors, dyes, organometallics, etc.) have been used as linearly absorbing media for optical limiting. Although various limiting mechanisms have been followed to achieve the optical limiting in different materials, these nonlinear mechanism-based optical limiting devices can be broadly classified into two categories: energy spreading type optical limiters and energy-absorbing type optical limiters as depicted in Fig. 2.

Continuous wave

2.1. Nonlinear absorption

The high-intensity radiation makes the material behave non-linearly by changing the real and imaginary parts of the polarization of material nonlinearly. Instantaneous and accumulative nonlinearities contribute to the nonlinear absorption as per their induced polarization density. For instance, accumulative nonlinearities in contrast with instantaneous nonlinearity depend strictly on the amount of energy accumulated into the medium and fluence of incident pulse and are therefore insensitive to pulse duration. Excited-state absorption, reverse saturable absorption, nonlinear refraction, and free-carrier absorption processes are such kinds of nonlinearities playing a vital role in optical limiting processes. On the contrary, instantaneous nonlinearities when incorporates for optical limiting processes, depict a broadband response but for short optical pulses mainly. Two-photon absorption (TPA) and multiphoton absorption (MPA) fall into the category.

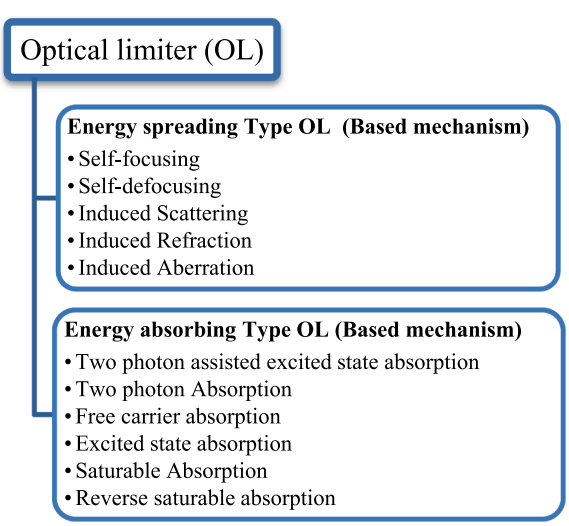

Fig. 2. Categorization of Optical Limiter based on nonlinear mechanism.

2.1.1. Reverse saturable absorption

A populated excited state on light absorption by the material contributes to reverse saturation absorption (RSA). In contrast with saturable absorption where transparency of material increases with an increase in intensity due to ground state depletion, RSA enhances the absorption of high incident laser intensity due to a large absorption cross-section of excited state than absorption cross-section of the ground state. At high input intensity, the ratio of absorption cross-section of excited state to absorption cross-section of the ground state should be large for minimizing output transmission. Few dyes [27–29] and Fullerenes [30–33] possess RSA phenomena to achieve optical limiting.

2.1.2. Multiphoton absorption

Multiphoton absorption (MPA) involves several virtual intermediate states for the simultaneous absorption of multiple photons. For short-pulsed (sub nanoseconds to few picoseconds) high-intensity radiation, MPA including two-photon absorption is scrutinized to be a quite efficient nonlinear technique for optical limiting. Two-Photon absorption (TPA) incorporating instantaneous nonlinearities provide broadband IR optical limiting for short optical pulses (less than nanoseconds). TPA is more advantageous over other nonlinear absorption (NLA) processes such as induced refraction, reverse saturable absorption, nonlinear scattering, etc. for optical limiting. This happens due to their very less linear absorption and very quick response to input intensity variations by offering donor-acceptor sites. Multiphoton absorption and excitation processes have been considered attractive over the past many years especially in semiconductor materials [34–37] and chalcogenide glasses [38–40]. To date, a very high order MPA coefficient up to 11th order has been demonstrated for GeSbS chalcogenides [41]. In AsP, InSb, InAs, HgCdTe, InGaAs semiconductor materials, multiphoton absorption rises with a reduction in bandgap values [42]. It is worth mentioning that these peculiarities are also the characteristics of other semiconductors, in particular, ABQ_2 (A = Ag, Li; B = Ga, In; Q = S, Se) and $ZnGeP_2$ [43,44]. These chalcogenide semiconductors possess useful NLO properties for nanosecond and femtosecond NIR regimes, inducing them as alluring optical limiting materials. Instantaneous and efficient protection against high laser intensity and linear transmission for ambient intensity makes MPA processes suitable for optical limiting processes.

2.2. Nonlinear refraction

When high-intensity radiations interact with nonlinear material, the index of refraction of the medium changes nonlinearly. Consequently, the incident intensity either focuses or defocuses the material as per the positive or negative refractive index variation [5]. This variation arises the self-focusing and self-defocusing nonlinear outcomes, which can further be helpful in optical limiting (OL) of materials. In both cases for high irradiance, the output intensity gets reduced due to the spreading of energy in the aperture plane. Self-defocusing is preferred over the self-focusing mechanism for optical limiting as self-focusing can damage the material due to intense field at the focal spot [45]. For CW laser radiations the defocusing mechanism is usually analogous to thermo-optic originated non-linearity which can be used to fabricate an optical limiter. Phthalocyanines, porphyrins, ferrosols, etc. materials have been examined for their nonlinear refraction-based optical limiting response [46–49] (see Table 1).

Table 1
Limiting materials and nonlinear mechanisms.

Materials	Mechanism	Transmission band
Carbon nanotubes	Nonlinear Scattering	Visible-NIR
Semiconductors and Chalcogenides	Multi-photon absorption Nonlinear Scattering	IR
Fullerenes and Dyes	Nonlinear Absorption Nonlinear Refraction	IR
Carbon black suspension	Nonlinear Scattering	UV-NIR
Metal clusters and Semiconductor nanocomposites	Nonlinear Scattering, Free carrier absorption	Visible-NIR
Graphene and derivatives	Nonlinear scattering, Nonlinear absorption	Visible-NIR
Organometallics	Reverse saturable absorption	Visible
Nanodiamonds	**Nonlinear Absorption Nonlinear Refraction**	**Visible-NIR**
Nanowires	**Nonlinear Scattering Reverse saturable absorption**	**Visible-IR**

2.3. Nonlinear scattering

Nanoparticles of semiconductors and metals, nanoscale carbon materials, and suspended carbon nanotubes depict the optical limiting mechanism via nonlinear scattering [50,51]. The nanoparticles and suspensions absorb the incident laser radiations, get heated up, and start vaporization. Scattering centers are created by heated up the microplasma. Thus, created microplasma further heat the surrounding solvent by thermal conduction and form the vapor bubbles. As the incident laser radiation intensity gets hiked, the scattering centers and bubbles formation gets vigorous and nonlinear scattering, as a result, gets a significant increase. As per the period of scattering center formation, direct bandgap semiconductor nanocomposites show optical limiting for subnanosecond pulses, indirect bandgap semiconductor nanocomposites limiting ranges from picosecond to hundred nanosecond laser pulses, whereas CBS is best suited for pulses longer than 10 ns and not for picosecond laser pulses [52]. The transmission band of these nanocomposites and suspensions ranges from UV to IR region. Perovskite type [$SrBi_2Ta_2O_9$ (SBT) and $PbZr_xTi_{1-x}O_3$ (PZT)] ferroelectric thin films follow the nonlinear scattering of ferroelectric domain walls mechanism for their optical limiting response [57]. Yang et al. [57] investigated their nonlinear optical properties towards 8 ns laser pulses at 1.064 μm and reported the limiting threshold as 5.84 J/cm^2 and between 4.53 and 5.93 J/cm^2 respectively and determined the damage threshold as 10.0 J/cm^2 and between 10.37 and 10.54 J/cm^2 for SBT and PZT respectively. A wide variety of carbon black suspensions, CNT suspensions, metal composites have been investigated for their nonlinear scattering technique-based optical limiting properties [53–57].

3. Optical limiting materials

3.1. Fullerenes and its derivatives

It is evident by recent investigations that Fullerene cages and their incorporated derivatives are proving very good OL materials with the best effectiveness in the IR wavelength range [58–64]. Such an optical limiting behavior is depicted in Fig. 3, where a comparative limiting response of parent C_{60} and its derivative is shown for different transmittance values. Undoubtedly nonlinear absorption exhibits a key role in limiting the response of fullerenes and their derived polymers [30–33]. Solid host C_{60} derivatives limit transmission by nonlinear absorption only but in liquid limiters, the mechanism is supplemented by multiple nonlinear mechanisms [65–70]. Brandelik et al. [71] demonstrated the comparative limiting behavior of the C_{60}/C_{70} mixture to CBS, reporting the 330 mJ/cm^2 limiting threshold of C_{60}/C_{70} in the toluene

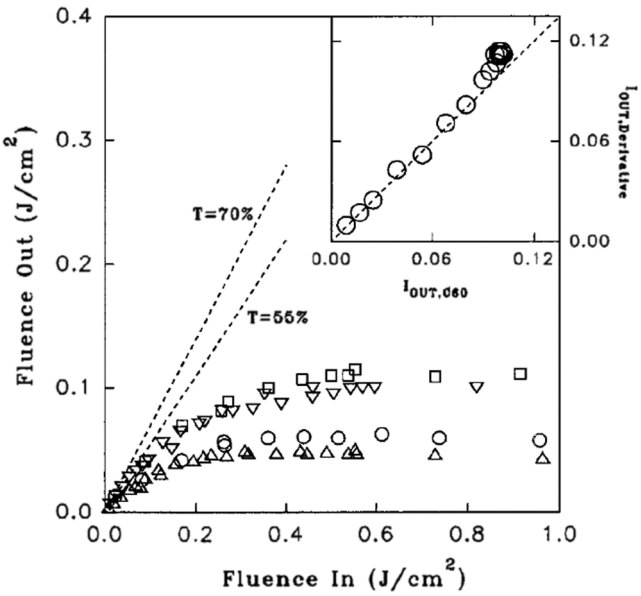

Fig. 3. The optical limiting behavior of C_{60} and methano-C_{60} derivative I in toluene solutions for linear transmittances 55% (C_{60} (Δ); I (○)) and 70% (C_{60} (▽); I (□)) at 532 nm. In the inset, I_{out} variation of C_{60} derivative is plotted against parent C_{60} for same I_{in} and same 70% linear transmittance at 532 nm. Reprinted with the permission from [73] Ya-Ping Sun et al. J. Phys. Chem. A, Vol. 103 No. 4, (1999) © American Chemical Society (with original caption).

solution for 66% linear transmittance and 19 mJ/cm^2 threshold for 34% transmittance. Medium viscosity creates a dramatic difference in OL behavior of fullerene derivatives in solution and solid-state polymer matrix host. For instance, Kost et al. [72] demonstrated the optical limiting behavior of C_{60} derivative in a polymethylmethacrylate (PMMA) matrix poorer than that in the toluene solution but Riggs and Sun [73] presented the increasing optical limiting performance of solid-state matrix over a solution of fullerenes. Tang et al. [74] carried out the investigation that improved σ_T/σ_G ratio make polymer-bound fullerenes (C_{60}-polycarbonate, C_{60}-polystyrene, and C_{60}-polyvinylchloride) perform better optical limiting response as compared to their parent fullerene for 532 nm excited 8 ns laser pulse.

Vincent et al. [75] came up with the idea that the parent C_{60} presents better OL performance with threshold value 0.1 J/cm^2 and 50% transmittance for 532 nm laser than its fullerol, alcohol, and amine derivatives. Kamanina and Sheka [76] studied the intermolecular interaction of composite film of fullerenes and COANP molecule (2-cyclooctaylamino-5-nitropyridine) and observed that the incident radiation got reduced two-fold as compared to parent fullerene for COANP + C_{60} film with 5 wt% of fullerenes for an incident energy density value of 1.5 J/cm^2.

3.2. Organic dyes

Sun and co-workers [77] reviewed recently a molecular engineering-based new approach of synthesizing supramolecular RSA (Supra RSA) for desirable optical limiting range. These super RSAs provide a better opportunity for improving the OL performance of organic dyes. The popularized intramolecular energy transfer techniques [77,78] are making their space in research for enhanced optical limiting behavior.

Organic dyes are actively investigated nonlinear absorptive materials [27–29]. Their fast vibration relaxation in electron excited states makes them suitable even for picosecond laser pulses limiting. Largely studied macrocyclic such as phthalocyanines, porphyrins, naphthalocyanines, and their derivatives [79–83] present the enhanced limiting behavior due to increased triplet yield and its lifetime. Shirk et al. [84] studied the limiting behavior of PbPc(β-CP_4) (lead tetrakis

β-cumylphenoxyphthalocyanines) solution in chloroform and observed a very low limiting threshold of 0.07 J/cm^2 for 8 nJ incident energy of 532 nm laser pulse. Brunel and co-authors [85] presented an improved threshold of 0.01 J/cm^2 with inorganic matrices doped with tetraphenyltetrabenzoporphyrines (PdTPTBP and ZnTPTBP). Perry and co-workers [86,87] incorporated the heavy metal atoms, and halogens in phthalocyanines and naphthalocyanines to enhance the optical sensitivity of limiters. The output energy got reduced to a factor of ten for an equivalent unsubstituted device and the transmission band got shifted by 100 nm for octabutoxy substituted naphthalocyanines as compared to the parent compounds. 3,5-dipyrenylvinylene BODIPY dyes are recently studied by Gugu et al. [88] with limiting threshold values below 0.95 J/cm^2 in the form of thin films at 532 nm excited wavelength laser. Incorporation of dyes in polymer hosts [89] or embedding into thin films make the limiter more stable with promising limitation of chemical compatibility a few times.

3.3. Carbon black suspension

Carbon black suspension (CBS) is considered to be the best optical limiting material due to its broadband energy limiting character and cheap availability of CBS solution for limiting more than 10 ns laser pulses [52,90]. Since 1989, CBS has been extensively studied for its excellent limiting behavior [91]. Nonlinear Scattering triggers optical limiting in CBS by creating microplasma and bubbles in the surrounding solvent due to the heating effect induced on excitation with intense laser radiations. Scattering centers thus formed in different solvents depict different limiting effects [92]. Belousova and co-workers [93] investigated the nonlinear optical limitation in CBS and fulleroid nanoparticles over a wide wavelength range of 300–1064 nm. A comparative study of astralenes (suspension of fulleroid nanoparticles) in an aqueous solution could limit the radiation two times as efficiently as CBS particles do. At 532 nm laser pulse, a limiting threshold value of 0.15 J/cm^2 was obtained for CBS particles, having almost the same results for 1064 nm laser pulse, evidencing the broadband limiting possibility. The dependency of CBS limiting response on some elements such as the energy of vaporization, viscosity, and surface energy, etc. of solvent was summarized by Sun [23]. McEwan et al. [53] supported the idea with 14 different fluid samples of CBS under different conditions and observed that the fluids with lower viscosity, surface tension, and vaporization show higher sensitivity. The chloroform-based suspension has a very low limiting threshold of value of 0.1 J/cm^2 [53]. Particle size role was reported by Vincent in his work, where smaller particles of 100 nm size in a suspension showed a better response in comparison to 200 nm size suspension [94]. Indeed, the limiter also represented the broadband limiting in the UV-NIR region with large stability.

3.4. Carbon nanotubes suspensions

The importance of CNT suspensions comes from the fact that they are the combination of relative advantages of fullerene solutions as well as of CBS. Like fullerenes, their electronic and structural properties are characterized well, and their large particle size suspension is comparable to CBS. Since the invention of CNTs by Iijima in 1991 [95], their fascinating optical properties have captured the attention of researchers [96,97]. The OL mechanism in CNTs is dominated by nonlinear scattering because of thermally induced micro bubble growth and carbon nanotube sublimation [54,55]. Sun et al. [98] in 1998 first carried out the study on multiple wall carbon nanotube suspension in distilled water and implanted in polymethylmethacrylate (PMMA) with the best limiting threshold value of MWNT's suspended in water as 1, 5, and 13 J/cm2 at 532 nm, 700 nm, and 1064 nm laser pulses respectively as presented in Fig. 4. Vivien et al. [99] explored the single-wall carbon nanotube (SWNT) suspension in water/surfactant solution and compared their limiting performance with C_{60} and CBS in the visible and NIR region. Adding to the same, Vivien and co-workers [51] evaluated

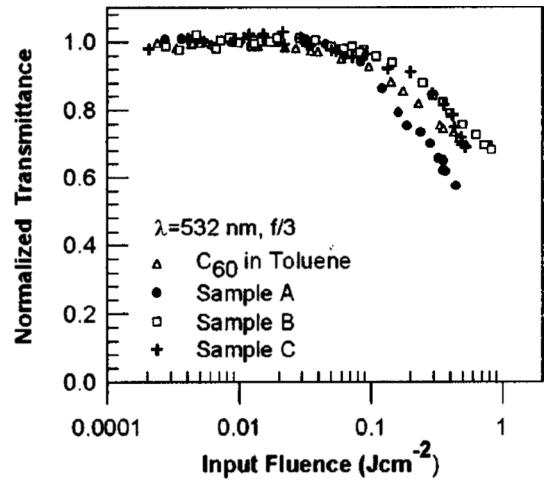

Fig. 4. Nonlinear transmission of MWNTs in water (sample A), MWNTs in PMMA (sample B), MWNTs, and carbon particles in PMMA (sample C) and C_{60} in Toluene measured for 532 nm nanosecond pulsed laser. Reprinted with permission from [98], X. Sun et al. App. Phy. Lett. Vol. 73, No. 25, (1998) © American Institute of Physics (with original caption).

the threshold values of SWNT in water as 150 mJ/cm^2 and 350 mJ/cm^2 towards 5 ns laser pulse at 532 nm and 1000–1064 nm excitation wavelength respectively with 24% and 58% transmittance. Also, SWNT in chloroform represented the threshold and transmittance value of 40 mJ/cm^2, 13% and 150 mJ/cm^2, 42% towards 532 nm and 1000–1064 nm, 5 ns laser pulses respectively, proving chloroform suspensions very good candidate for limiting for long pulses also [51].

Nanotube suspensions exhibit quite different optical limiting than functionalized CNTs. Like CBS, CNT suspensions limit the high intensity via the NLS mechanism while functionalized CNT relies on the absorption mechanism to form homogeneous solutions for OL. For instance, the limiting response of (Polypropionylethylenimine-co-ethylenimine) (PPEI-EI) polymer-bound SWNTs in chloroform solution of 70% laser transmittance is considerably lower than that of aqueous SWNT suspension at 532 nm [100]. Functionalized MWNTs supported the same weaker OL response than suspended MWNTs for 532 nm laser excitation [100]. Despite these contradictory limiting of suspension and solution phase CNTs, these systems are very advantageous in recognizing the limiting mechanism and phenomena for effective optical limiting.

3.5. Nanodiamonds

Very little is studied about the optical nonlinearities of nanodiamonds, a kind of carbon nanocomposites. Diamond belongs to group IV semiconductors, which has a four-fold coordinated structure exhibiting a well-known covalent sp^3 bonding. This type of structure arises a larger gap (>5 eV) in the state of its electronic density, complementary to the nanoscale diamond which associates with the three-fold coordinated atoms showing sp^2 covalent bonding. Such behavior of nanodiamonds enhances their NLA and hence OL response in comparison to diamond itself. Very less but quite significant work has been performed by few research groups past decade to explore the nonlinearities of diamond nanocomposites and functionalized nanodiamonds [101–104]. Papagiannouli et al. [102] prepared the crystalline nanodiamonds (NDs) and amorphous carbon dots (CDs) and studied their third Order NLO response for 532 and 1064 nm, 35 ps, and 4 ns laser pulses with a repetition rate of 1–10 Hz. No significant difference was observed at different repetition rates. However, in ps and ns regime and IR excitation, CDs exhibited stronger NLR and negligible NLA while NDs showed self-defocusing both for visible and IR excitation. The NLO response of CDs was found two orders of magnitude higher than NDs. Though NDs also exhibited a significantly lower limiting threshold of values 0.25 J/

cm^2 and 0.5 J/cm^2 for 532 and 1064 nm excitation wavelength, 4 ns pulsed laser respectively. Onion-like carbon and diamond nanoparticles are also a kind of carbon nanocomposites, which are attracting the attention of researchers in the past few years [105,106]. Koudoumas and colleagues [107] investigated the limiting behavior of ultradispersed and polydispersed diamond powders along with onion-like carbon in water suspension under 532 nm excitation wavelength and 10 ns pulsed laser as shown in Fig. 5. Dominating NLA in onion-like carbon structures exhibited their strongest OL in comparison to diamond nanoparticles.

The densest allotrope of carbon i.e. nanodiamond is a structure where atoms are arranged in a tetrahedral close packing through sp^3 covalent bonds. A detailed investigation of such surface-functionalized nanodiamonds (DNDs) was performed by Muller and coworkers [101]. Amino-terminated DNDs (DND-NH_2) and hydrogen-terminated DNDs (DND-H) were compared to unfunctionalized DNDs for their nonlinear optical mechanism and hence their optical limiting response. Their calculated optical threshold values were 65 J/cm^2 for the DND system, 25 J/cm^2 for DND-NH_2, and 15 J/cm^2 for DND-H for 50% transmittance drop. NLR with significant NLA predominant in DND-H suspension instead of NLR only that occurs in unfunctionalized DNDs. For future perspectives, diamond nanocomposites, functionalized nanodiamonds, and their thin films are a long way to go for their NLO response and OL behavior.

3.6. Semiconductor and other metal nanocomposites

Nanoscale semiconductors and metal clusters have been investigated as optical limiters for the past few years [108]. Using the supercritical fluid method, Ni metal nanoparticles [109] and silver and silver sulphide nanoparticles in PVP polymer suspensions [110] were prepared and investigated for their optical limiting behavior. Unlike the NLS mechanism in nanoparticles, the optical limiting here is prevailed by the nonlinear absorption mechanism based on the spectroscopic results. Silver sulphide nanoparticles with 7.3 nm average size depicted the best optical limiting response over other metal sulphide nanoparticle suspensions under 5 ns laser pulse at 532 nm wavelength. The particle size dependency of optical limiting was reported by Francois and co-workers [56], where they recorded the response of gold nanoparticles of size 2.5, 9, and 15 nm radii in water solution and polyvinyl alcohol (PVA). Increasing particle size decreases the fluence threshold and its amplitude is increased, dominated by nonlinear scattering. The largest gold particle solution represented the best optical limiting towards 530 nm laser pulse. In semiconductor nanoparticles, the optical limiting mechanism is

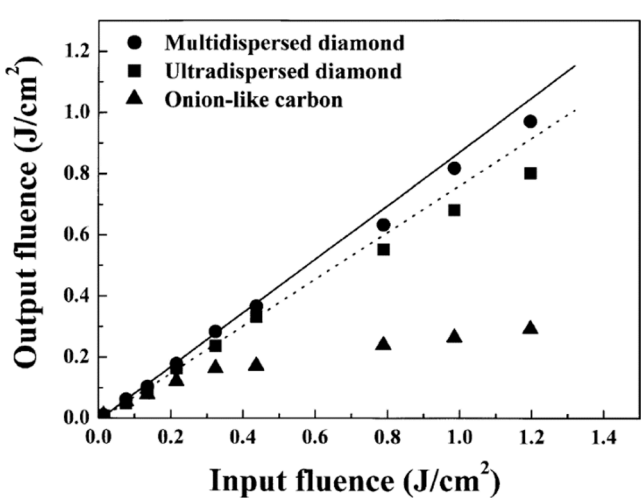

Fig. 5. Optical limiting of the ultradispersed diamond (■), the multidispersed diamond (●), and the onion-like carbon (▲). Reprinted with permission from [107], E. Koudoumas et al. Chem. Phy. Lett. Vol. 357, (2002) © Elsevier Science (with original caption).

contributed by nonlinear scattering as well as by free carrier absorption (FCA).

Jia and co-workers [111] claimed that the OL performance of $Cd_xAg_{1-x}S$ semiconductor nanoparticles was increasing with increasing nanoparticle size along with increased FCA. OL of Ag_2S thin layer coated CdS particles also got enhanced for a minimum threshold value of 0.4 J/cm^2 for CdS/Ag_2S nanoparticles for 4.1 ns pulsed laser of 532 nm wavelength. Effective optical limiting in metal and semiconductor nanoparticles is still to be explored. For evaluating the potent OL, explicit fabrication and characterization of nanoscale particles is a future challenge.

3.7. Organometallics

Organometallics with compatible nonlinear photonic applications are consisting of one metal at the least. Extra metallic element in organometallics makes them superior over organic compounds by adding on many more optical transitions. Indeed, for transition metals, even electronic transition emerges from ligand (organic portion) to metal (LMCT) or from metal to ligand (MLCT) as per the situation of electron donor, while in organometallics intraligand transitions take place. Metal cluster compounds and metal macrocycles are two extensively studied categories for optical limiting applications under the RSA mechanism [112–118].

Metal clusters consisting of two or more than two metals are stabilized compounds due to their ligand satisfied metal valency. King's complex, a group of tetra iron compounds (Fe (CO_4) cyclopentadienyl) with parent complex was reported on its first synthesis by RB king [112]. Boggess et al. [113] investigated cyclopentadienyl iron carbonyl tetramer (King's complex) for picoseconds and nanosecond pulses of 532 nm excitation wavelength. They demonstrated that the OL in King's complex solution is influenced by thermally induced scattering while the polymethylmethacrylate (PMMA) solid host incorporated King's complex exhibited negligible induced scattering. Nonlinear absorption in (n-$Bu_4N)_4$ ($Mo_8Cu_{12}O_8S_{24}$) cluster compound depicted a threshold value as lesser as 0.7 J/cm^2 for 7 ns pulsed 532 nm laser irradiance [118].

King's complex solution in methylene chloride [114] has been investigated for nanosecond time scale pulsed OL. Metal phthalocyanines and naphthalocyanines metal macrocycles exhibit two types of absorption bands [115], in the UV region its Soret band, and in IR its Q bond. RSA process dominates the optical limiting in these materials at 532 nm pulse. Silicon naphthalocyanines (SINC) investigated by Perry et al. [117] for broadband limiting, metal-free tetraphenyl porphyrins (H_2TPP), ZnTPP and CoTPP metal complexes reported by Blau and co-workers [78] and metal clusters demonstrated by Ji and co-workers [118], all these metal macrocycles possess RSA and represented the optical limiting dynamics.

Due to attractive featured nonlinearities, organometallics have been successful to earn a place in the optical limiter family. Further efforts can be performed to achieve its limiting threshold to much lesser values.

3.8. Semiconductors

Semiconductors are considered to be one of the best materials exhibiting a huge diversity of nonlinearities. For optical limiting in semiconductors, a large variety of nonlinearities such as TPA, Kerr effect, nonlinear refraction along with free carrier generation, and free-carrier absorption has been extensively studied [119–122]. TPA associated with free carrier generation was firstly reported by Ralston and Chang [123] in CdS, CdSe, and GaAs semiconductors for optical limiting applications. TPA associated with nonlinear refraction was demonstrated by Walker and co-workers [124] to limit the 10 μm wavelength laser irradiation but with a drawback of smaller input energy limiting range.

Van Stryland and his co-workers have investigated a large array of semiconductors [125–127] for absorption as well as refractive

nonlinearities. Indeed, ZnSe being their extensively studied semiconductor for TPA-based optical limiting applications, represented broadband limiting similar to GaAs with a short pulse requirement for effectual limiting. Thomas F. Boggess et al. [128] were the first to emphasize an integrated effect of TPA and free carrier defocusing for optical limiting in GaAs semiconductors. They complemented the TPA with nonlinear refraction and phase transition due to laser induction to enhance optical nonlinearities of GaAs semiconductors. The smallest value of the TPA coefficient ($\beta = 26 \pm 8$ cm/GW) was reported with limiting the capability of limiter up to 1.2 mJ of input influence for picosecond pulse of 1 μm excitation wavelength laser. L. W. Tutt [5] reviewed the importance of Si semiconductor (indirect bandgap semiconductor) with a 1.1 eV bandgap over narrow bandgap semiconductor for laser intensity of 1.06 μm excitation wavelength. Boggess et al. [129] explained the fluence dependent nonlinearities of Si in term of the number density of generated carriers by nonlinear refraction and absorption mechanism for 25 ps pulses of 1.06 μm wavelength. A massive amount of carrier densities is generated because of a very high density of states in the valence and conduction band, which alter the refractive index and lead to a prominent limiting behavior. GaAs in contrast to Si [128] being a direct bandgap semiconductor represented an intensity-dependent optical limiting. Indeed, Si and GaAs operate well for the different limiting mechanisms, but their bandgap energy values do not fit them for visible range optical limiting. However, in past few years, a wide range of semiconductors have been investigated which fits suitable for broadband optical limiting.

Nonlinear refraction and nonlinear absorption can be combined to focus the laser irradiance into semiconductors inventing a self-protecting optical device that operates for a value much lesser than its damage threshold value. ZnSe semiconductor [125] depicts this self-projecting behavior by a combining mechanism of TPA and free carrier self-defocusing over a broadband range. This device limited input laser intensity up to 10 nJ (300 W) for 0.53 μm excitation wavelength and 30 ps pulse duration over a dynamic range of more than 10^4 (see Table 2).

Guang et al. [130] demonstrated the multiphoton excitation and multiphoton absorption properties of varying size quantum dots of CdSe in hexane with 160 fs pulsed laser of 775 nm and 1300 nm excitation wavelength. Before that only the TPA mechanism was reported for semiconductor quantum dots (QD). However, CdSe QD solution in hexane with two and three-photon absorption and excitation properties has emerged as new possibilities of inorganic semiconductor QDs in intense laser power limiting applications with tremendous capabilities of large photo-physical and photo-chemical stabilization. Guang [131] extended his work by studying CdTe quantum dots in chloroform. The feasibility of OL for 160 fs laser pulse of 1300 nm wavelength (widely used in optical communication) with superior limiting performance for 10 μJ input intensity was reported. The nonlinear transmission got reduced from 100% to 20% for a variation of 50 nJ to 10 μJ input pulse energy. Recently graphene nanocomposites along with semiconductors have also attracted a lot of interest for optical nonlinearities. GO-ZnO/TiO_2 semiconductor nanoparticles have been investigated [132] for their size and shape-dependent optical properties arising from NLS and TPA processes. Semiconductor nanocomposites exhibited the higher energy absorption shift induced by quantum confinement leading to broadband optical limiting applications. The limiting threshold of GO, GO-ZnO, and GO-TiO_2 composite were estimated to be 0.80, 0.48, and 0.55 J/cm^2 respectively at 532 nm wavelength and 1.65, 1.15, and 1.35 J/cm^2 at 1064 nm.

Though these semiconductor-based optical limiters exhibit a wide variety of aspects, yet to achieve an OL closer to an ideal optical limiter, lots of features such as bandwidth along with intensity-dependent mechanisms are still required.

3.9. Chalcogenides

Limiting photonic applications of chalcogenide glasses solely relies on their high order NLO properties (two orders of magnitude greater than silica) and broader transparency window spanning from NIR to 20 μm. Their ability to exist in variant forms such as bulk, thin films, nanosheets, or fiber makes them suitable for lots of tremendous classes of photonic applications [133–136]. Now a day, chalcogenides thin films [137–139] have fascinated the considerable attention of investigators for their nonlinear optical parameters. These parameters depend on growth techniques and other external elements. Ganeev et al. [140] demonstrated the optical limiting of chalcogenide films $As_{20}S_{80}$, As_2S_3, $3As_2S_3/As_2Se_3$, and $2As_2S_3/As_2Se_3$ for a picoseconds laser of 1064 nm and 532 nm excitation wavelength. Their work elucidated that the free carriers generated due to TPA alter the linear absorption and refraction of the material. TPA coefficients and large nonlinear absorption coefficients were measured and the limiting process was observed in the laser intensity range 10^7 to 10^9 W/cm^2. The author characterized the limiting of $As_2Se_3/2As_2S_3$ films as 12.5-fold and As_2S_3 as 25 fold. Ganeev with his co-workers [119] studied the different limiting mechanisms of picoseconds and nanoseconds laser radiations for fullerene solutions, colloidal metal solutions, and semiconductor crystal structures. A theoretical analysis of optical limiting in a bunch of chalcogenide films was carried out for fundamental wavelength 1064 nm and its second harmonic 532 nm. $As_2Se_3/2As_2S_3$ films attenuated the signal by a factor of 12.5 with the best efficiency, followed by As_2Se_3 and

Table 2
Limiting properties of nonlinear materials and devices.

Material	Incident wavelength (nm)	Pulse Duration	Repetition Rate (Hz)	Transmission (%)	Limiting Threshold (J/cm^2)	Reference
SWNT (Chloroform)	532	5 ns	–	13	0.04	[51]
PbPc(β-CP)$_4$/CHCl$_3$	532	8 ns	10	62	0.07	[84]
SWNT (Water)	532	5 ns	–	24	0.15	[99]
Aq GO	532	35 ps	10	70	0.2	[165]
C_{60} (Toluene)	532	5 ns	0.2–10	65	0.2	[58]
Gold (Water)	532	14 ns	1	–	0.2	[56]
Methano-C_{60} Benzoyl (Toluene)	532	5 ns	0.2–10	70	0.35	[58]
Ag NPs/rGO	532	10 ns	10	80	0.38	[164]
(n-Bu$_4$N)$_4$[Mo$_8$Cu$_{12}$O$_8$S$_{24}$]	532	7 ns	2.5	–	0.7	[118]
CBS (Water)	532	20 ns	0.2	40	1	[53]
Gold NWs	**532**	**7 ns**	**1**	**70**	**1.56**	[170]
WSe_2	532	6 ns	2	55.1	7.2	[145]
Graphene	532	6 ns	2	79.7	15.15	[145]
Sb_2Se_3-GO	**1064**	**5 ns**	**10**	**50**	**0.085**	[176]
CBS	1064	10 ns	10	–	0.5	[90]
Aq GO	1064	35 ps	10	96	0.60	[165]
$PbZr_xTi_{1-x}O_3$ (PZT)	1064	8 ns	–	–	4.53–5.93	[57]
$SrBi_2Ta_2O_9$(SBT)	1064	8 ns	–	–	5.84	[57]
$MoSe_2$	1064	6 ns	2	25.3	9.8	[145]

$As_{20}S_{80}$ attenuating by a factor of 5 and $As_2Se_3/3As_2S_3$ films attenuating by a factor of 3 in the intensity range 10^7 to 10^9 W/cm^2.

Three glasses from Ge-As-Se ternary Chalcogenide system and two glasses from the PbI_2-As_2S_3-Sb_2S_3-Bi_2S_3 family [141] exhibited optical limiting efficiency comparable to fullerenes and a little lesser than CNT suspension at 1.064 μm wavelength. Out of several Chalcogenide glass families investigated by Troles et al. [142], As_2Se_3 and $GeAs_2Se_2$ displayed the best OL performance, $GeAs_2Se_2$ with a transmittance value of 0.65 GW/cm^2 in place of 3GW/cm^2 for 5 GW/cm^2 incident intensity while $(GeSe_4)_{1/2}$ $(AsSe_3)_{1/2}$ reduced the input intensity to 1.5 GW/cm^2, showing the least efficiency.

Now a day, inorganic or polymeric nanocomposites are emerging as advantageous materials with their unique physical and optical properties. Tintu and co-workers [143] for the first time reported the $Ge_{28}Se_{60}Sb_{12}$/PVA thin film fabrication and their optical characterization. The colloidal solution was prepared by the dissolution of $Ge_{28}Se_{60}Sb_{12}$ glass in n-butylamine solvent for different concentrations followed by a drop-casting method to deposit the films. The best optical limiting response noted was 423 MW/cm^2 transmitted power density corresponding to 536 MW/cm^2 input power density for 1200×10^{-3} mg/ml concentration sample. Indeed, the increased cluster size depicts the least limiting threshold and best-limiting performance.

Nonlinearity demonstration of chalcogenide nanosheets of MoS_2, NbS_2, and WS_2 by Zhou et al. [144] for ps pulses at 532 nm excitation wavelength enlightens the optical limiting path of these materials for future research. Dong and co-workers [145] made a comparative study of optical limiting of transition metal dichalcogenides nanosheets (TMDC) with graphene. The author recorded the strong limiting behavior of TMDC and graphene from the NLS process for NIR laser while TMDC represented a combined effect of NLS and SA for 532 nm laser. For 6 ns pulsed laser intensity and 1064 nm excitation wavelength $MoSe_2$ exhibited the best limiting response with the threshold value of 9.8 J/cm^2. Lower limiting threshold values and minimum nonlinear transmittance value makes TMDC a potential candidate for them to work effectively as an optical limiter in NIR as well as in visible range.

Similar to TMDCs, 2D transition metal dichalcogenides (2DTMDCs) have also paved the way for OL applications [146–149]. Few TMDCs such as TiS_2 [146], MoS_2 [147], $MoSe_2$ [148] and WS_2 [149], etc. exhibit optical limiting effects due to their giant TPA coefficient, which is attributed to 2D confinement of electrons, RSA, band-edge resonance, and giant exciton effect [149]. The number of layers and excitation wavelength also plays a key role in the OL of 2DTMDCs along with the input fluence because of layer-dependent bandgap [147,148]. Under NIR excitation wavelength, TPA dominates over SA because of the larger bandgap of the monolayer to a few-layer (less than4 nm thickness) of 2DTMDCs [147,148]. R. Wei et al. [147] synthesized successfully the ultrathin MoS_2 nanosheets (UMS) with <4 nm thickness and measured RSA and NLR response of UMS for 800 nm excitation wavelength and fs laser pulses with a limiting threshold value of ~44 mJ/cm^2. Recently Varma and co-workers [146] observed the larger OL response of TiS_2 nanosheets from one to two layers for 532 nm excitation wavelength in ns regime. TPA and induced ESA attributed to the larger modulation depth of OL response of TiS_2 nanosheets.

Layered thin films of chalcogenide glasses are an emerging field for the future. Due to their ultrafast nonlinearities and NLO properties, these are proving themselves important materials for upcoming optical limiters. These materials are yet to move far ahead in the optic field for future photonic applications and optoelectronic devices.

3.10. Graphene and its derivatives

Graphene has captured the focus of researchers due to its intensive nonlinear optical properties and wide-spreading applications in optical limiters [150–152]. An sp2-hybridized single-layered thick sheet of carbon atoms constitutes the honeycomb lattice of graphene. But its hydrophobic nature has created chromatic challenges for researchers restricting its use in aqueous media. Graphene's oxidized face i.e. graphene oxide (GO) has provided an alternative precursor for various graphene-based nanocomposites and nanosheets. The oxygen group abundance in GO makes them suitable for their chemical functionalization. To enhance the NLO properties of GO, various multifunctional groups as dye molecules, organic materials, and dielectrics have been attached to fabricate reduced graphene oxide, graphene oxide colloids, and graphene hybrids, etc. [153–155]. Wang et al. [156] reported the fabrication of monolayers of graphene and multilayer of graphene dispersion in an organic solvent, making it to exhibit unique nonlinear optical properties and broadband OL for ns pulses. Following the same trend of CNTs, and CBS [53,54,96,97], graphene suspension also ascribed the NLS mechanism for their optical limiting effects. However, recently NLA has also emerged as an effective mechanism [150,153,155] in the optical limiting of graphene-based nanocomposites. Feng and co-workers [157] demonstrated the broadband optical limiting of a large family of graphene including graphene nanosheets, GO nanosheets, graphene nanoribbons, and GO nanoribbons for 532 nm and 1064 nm excitation wavelength ns pulses. Amalgamate of NLS and TPA mechanism established the strong optical limiting in graphene nanostructures. Indeed, the GO precursor revealed the weaker OL response in comparison to rGO due to enlarged crystallinity and conjugation of rGO. Balapanuru and co-workers [158] reported GO-dye charge transfer synthesized by the ionic exchange process and exhibited the enhanced optical limiting and bio-sensing properties. Dong et al. [159] investigated a carbon nanotube GO core-shell structure formation due to the strong Π-Π stacking interaction of these CNTs and GOs. Ultrafast energy transfer or e^- transfer between two components suppressed the absorbed energy dissipation via photoemission and enhanced the OL response of SWCNT/GO as compared to their individuals for 532 nm, ns pulses.

GO when functionalized covalently with $NaYF_4$: Yb^{3+}/Er^{3+} nanoparticles (GO-Er) represented a better TPA efficiency than organic molecules [160]. NLS effect in GO synthesized with upconversion of lanthanide ions Yb^{3+}/Er^{3+}in $NaYF_4$ nanocrystal exhibited the optical limiting over a broadband range for 800 nm laser pulses. Wei with his co-workers [161] also represented the GO-URENs (Go-upconversion rare-earth nanomaterials) of GO and $NaYF_4$: Yb/Er nanocrystal by combining the benefits of both GO and URENs for an enhanced optical limiting response. This upconversion widened the limiting range of OL from visible to NIR region along with its improved lightweight and high-performing capability. Zhang et al. [162] investigated the GO-Fe_3O_4, a hybrid material of graphene oxide and Fe_3O_4 individuals, synthesized by chemical precipitation technique. Fe_3O_4 nanoparticles were attributed to the NLS mechanism responsible for enhanced OL properties of 532 nm pulsed laser intensity over GO and C_{60}-benchmark material.

Tetra amino porphyrin-graphene oxide hybrid (GO-TAP) a covalently functionalized structure inorganic solvents were experimentally demonstrated by Yamuna et al. [163] for its optical nonlinearities and optical limiting behavior. The higher absorption coefficient of value 13.1×10^{-4} cm/W was reported for hybrid structure as compared to its individuals. A combined effect of TPA, RSA, and NLS is attributed to optical limiting applications. Recently, Yang [164] synthesized the AgNPs/rGO hybrid by laser ablation into the mixed aqueous solution of GO and $AgNO_3$. NLS along with NLA contributed to the optical limiting of hybrid material for 532 nm laser as compared to its GO component where only NLS was responsible for the OL process. A quite lesser threshold value of 0.38 J/cm^2 was depicted for 1 mM Ag ions of AgNPs/rGO hybrid. Liaros [165] mentioned the OL of aqueous graphene oxide (GO) dispersion for visible and IR wavelength under 35 ps laser pulses. GO suspension exhibited the OL for visible laser radiation comparable to C_{60} as well as for IR radiations making the limiter broadband with a limiting threshold value of 0.20 and 0.6 J/cm2 for 532 and 1064 nm laser respectively. 1.2 mg/ml aqueous GO suspension showed 70% linear transmittance at 532 nm and 96% at 1064 nm wavelength. Researchers of different backgrounds have put lots of effort into the better

tuning of more optical properties of graphene and their derivatives. Still, to bring the best out of graphene's ultra-wide capability and broadband limiting ability, scientists need to explore its high-speed device demonstration over a wide range of wavelengths.

3.11. Nanowires

1D nanostructured metals i.e., nanowires (NWs) are emerging as a vital player in novel imaging and sensing applications. In the past decade, significant work has been performed for exploring the non-linearity and OL behavior of metal nanowires [166–168]. The peculiar feature of quantum confinement of electrons exhibited by nanowires manifests itself in discrete energy levels instead of the traditional continuum of energy levels found in bulk materials. This quantum confinement is often referred to as specific non-linearity (NLS) that spans the optical limiting from visible to NIR wavelength range. Pan and co-workers [169] for the first time reported the OL of metal (Cu, Co, Ni, Pd, Pt, and Ag) NWs over a wide band of limiting capabilities. For nanosecond duration pulses, NLS attributed the major contribution to the optical nonlinearity of NWs. Anodic aluminum oxide (AAO) template was used to grow metal NWs (MNWs). At 532 nm wavelength laser, the optical limiting response of Pd, Ni, Pt, and Ag NWs were reported comparable to or slightly better than SWCNTs and MWCNTs with threshold values 0.9, 1.2, 1.3, and 1.7 J/cm^2 respectively. While Co NWs and Cu NWs showed slightly poorer optical response with threshold values 4.2 J/cm^2 and 2.5 J/cm^2 respectively. A similar trend of OL was presented by MNWs for 1064 nm excitation wavelength and the same 7 ns pulsed laser. Metal nanoparticles (NPs) exhibit their nonlinearity due to NLS and NLA but optical nonlinearity in MNWs is dominated by NLS, making them superior OL over metal NPs. S. Cai [170] claimed the benchmark threshold value of 1.56 J/cm^2 for ultralong gold NWs in water over CNT materials using 532 nm, nanosecond pulsed laser. Q. Chang and colleagues [171] evinced the enhanced limiting behavior in CdO NW suspension in ethanol over the C_{60} solution in toluene and CdO NW suspension in water due to stronger NLS mechanism.

II-IV group semiconductor, cadmium sulfide (CdS) has been investigated as CdS nanoparticles [172] and CdS multibranched and individual nanorods [168] for their nonlinear optical response. Muthukumar et al. [173] presented a detailed investigation of nonlinear optical transmission of CdS NWs. The size-dependent nonlinear performance was originated by effective three-photon absorption which was induced by single-photon absorption and TPA at 532 nm wavelength. As a result, despite the weaker quantum confinement, CdS NWs exhibited a threshold value comparable to Cd-based nano-structures [172]. In the same trend, Si semiconductor-based SiNWs are also emerging as a potential candidate for Si-based optical devices. Hui Pan and co-workers [174] have grown Si NWs using thermal vaporization technique on Si substrate. For 7 ns pulsed, 1064 nm laser, NLS dominated over RSA for the optical limiting of SiNWs. In contrast, for 7 ns pulsed, 532 nm wavelength laser, NLS and RSA both (due to Quantum confinement) contributed to better OL of SiNWs. Besides, Si-Si dimers with self-trapped surfaces played a vital role in better threshold value. For 70% linear transmittance, SiNWs exhibited a 5 J/cm^2 threshold value for nanosecond pulsed 532 nm laser. SiNWs depicted a remarkable OL for 532 nm laser over 1064 nm excitation wavelength laser.

Gao et al. [175] for the first time mentioned the solvent-dependent OL behavior of PbNWs stabilized by [60] fullerene derivative (C_{60} TPY-Pb). The optical nonlinearity of C_{60} TPY-Pb NWs measured in chloroform, ethanol, and dimethylformamide (DMF) for 532 nm excitation wavelength, 8 ns duration pulsed laser. In addition to absorption-induced NLS, heat capacity, thermal conductivity, boiling point, etc. factors also contributed to the OL of C_{60} TPY-Pb NWs. C_{60} TPY-Pb in chloroform exhibited the strongest OL with a least threshold value of 507 mJ/cm^2, while C_{60} TPY-Pb in DMF showed the weakest OL with 1300 mJ/cm^2 threshold value.

Nonlinear materials with the least limiting threshold (less than 1 J/cm^2) have always been in demand which can reduce the high-intensity transmission without sustaining a loss. CNTs have proved themselves the best candidate in this regime with a limiting threshold value of 0.9–1.1 J/cm^2 and damage threshold value of 1.5–3.5 J/cm^2. Nano-structured thin films of CuO NWs consisting of a mixture of Cu_2O and CuO phases [167] have competed with CNTs. With a limiting threshold value of 0.4 J cm^{-2} and a damage threshold value of 0.7 J/cm^2, these thin films have served as a benchmark in the direction of robust limiting device formation. An immense typical heterostructure of nanosheet-nanowires was successfully synthesized by RK Yadav [176] and colleagues for a giant increase in nonlinear absorption. Thus, formed Sb_2Se_3-GO heterostructure evinced an excellent OL over visible to IR region, setting a goal in OL onset of value 8 mJ/cm^2 and 20 mJ/cm^2 for 1064 nm and 532 nm excitation wavelength respectively which was better than transition metal dichalcogenides [145], metal nanoparticles [110], and graphene-based limiters [160]. For the same duration pulse, mentioned heterostructure exhibited better OL for 1064 nm excitation wavelength over 532 nm wavelength laser. Recently, Babeela et al. [177] mentioned that nanowires when decorated over nanoplatelets can exhibit micraculour OL action. The concentration-dependent NLA and resultant limiting response of Fe nanowires decorated nanoplatelets β-BaB_2O_4 was presented. Fe^{3+} ions when incorporated into β-BaB_2O_4 nanorods enhanced the NLA coefficient and reduced the limiting threshold over pure β-BaB_2O_4 nanorods. Highly concentrated 0.05 M Fe decorated β-BaB_2O_4 exhibited the least onset threshold value of 1.03 W/m^2 than 1.86 W/m^2 for 0.01 M Fe decorated β-BaB_2O_4. The recent work in metal nanowires and their fascinating OL performance in the wide regime of wavelength has served as a benchmark for future research work. The pace at which research in nanowires has been expanding makes us believe that many new physics pertaining to NLO response of nanowires are yet to be discovered (see Fig. 6).

4. 1-D photonic crystal

After the first introduction of a parametric subharmonic oscillator as an Optical Limiter in 1962 by Siegman [178], few other groups [179–181] also followed the same trend of oscillations as optical power limiters. Originally, Leite and co-workers [26] demonstrated a self-focusing mechanism for passive optical limiting. Subsequently, RSA in fullerenes, its derivatives, and in dyes, MPA in semiconductors and chalcogenide, NLS in CBS, CNT, etc. have emerged as an effective mechanism in a plethora of photonic applications past few decades. The

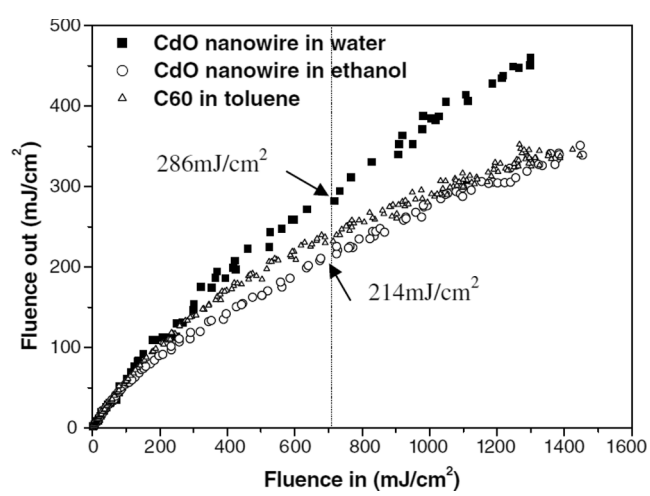

Fig. 6. The optical limiting response of CdO nanowire suspension in water (■), CdO nanowire suspension in ethanol (O), and solution of C_{60} in toluene (Δ) for the same 75% linear transmittance at 8 ns, 532 nm wavelength laser. Reprinted with the permission from [171] Q. Chang et al. Optics Communication Vol. 274, (2007) © Elsevier Science (with original caption).

simplest passive optical limiters mentioned to date are a single layer of nonlinear materials and their device design realization is provided by only imaginary part permittivity (ε'') dependency over input energy density. As explained by Makri [182], up to a certain lower intensity value, ε'' is very small and the material behaves linearly. Eventually, input intensity crosses the prescribed value also termed as the threshold value, ε'' increases large and the material becomes opaque. The objective behind this dependency can be TPA, RSA, NLS mechanisms, or their combination in different materials.

Despite the nonlinear mechanism implications for several optical limiting applications, a major problem arises with these single-layer devices. When these nonlinear layers are embedded in a complex optical setup, the nonlinear material absorbs the access laser intensity, gets overheated, and scarifies itself for protecting the device. Thus, a prior demand for self-protecting optical limiters along with a sophisticated optical device protector arises. Yoo and Alfano [183] approximated the mathematical modeling of a multi-layer system representing the nonlinearity depending on the uniform-intensity approach. Moving a step ahead Kahn [184] explored the dependency of OL and its threshold value over a variety of parameters. The approximation model was consisting of a multilayer structure having an alternate layer of slightly variant refractive index values. The transmission coefficient variation with input intensity and its wavelength was explained as an effect of Kerr nonlinearity. The dynamic range was expected to be layer thickness dependent.

Scalora et al. [185] numerically demonstrated an optical limiting of one-dimensional photonic bandgap (PBG) structure which was doped with Kerr nonlinearity. Optical limiting and optical switching were reported for ultra-fast pulses100 optical cycles long (around 300 fs for $\lambda = 1$). The multi-layer structure consisting of alternate layers of slightly different refractive index with $\lambda/4$ thickness each (λ is the design wavelength) formed a quarter-wave reflector, also termed as distributed Bragg reflector. λ centered wavelength range was reflected making the material a desired photonic bandgap structure for the optical limiting of a specified wavelength spectrum. A similar nonlinear finite-difference time-domain technique (FDTD) was investigated by Tran [186] numerically. He further extended his work by the introduction of a Kerr defect into a 1-D photonic crystal or a quarter-wave reflector [187]. He depicted the input intensity drop up to 80% in a reflector without defect and up to 85% in the reflector with a defect and managed to enhance the limiting and switching capability with a defect mode optical limiter.

A modulation instability-based transmission instead of traditional band edge shifting was investigated for homogenous as well as defected nonlinear OL by Soon et al. [188]. The optimization idea of OL performance by the selection of the material of photonic crystal (PC) and its geometry was represented for future work. They provided a set of physical parameters for guiding the experiments. For instance, for two materials of dielectric constant 0.01, nonlinear coefficient value 10^{-10} cm^2/W with a pulse width of 10 ns, the simulated threshold was approximately 3 mJ/cm^2 for a sample thickness of 12 μm. A variety of other parameters were also explored for more sophisticated PC optical limiters over the homogenous nonlinear OL for short pulses. 1-D photonic crystal, also known as planer microcavity resonators has emerged as the easiest exploitable photonic bandgap (PBG) device in the last few decades. The reason includes their much lesser cavity decay time over cavity mode-atom interaction time which can easily manage the rate of spontaneous emission of excited-state atom in a weak coupling regime. RF sputtering fabricated, high Q factor Er^{3+} activated microcavities [189,190] have been investigated for different Bragg reflector pairs of layers. Chiasera et al. [189] reported the 1-D photonic crystal cavity of Er^{3+} doped SiO_2 active layer sandwiched between Bragg reflectors of six pairs of SiO_2/TiO_2 layers. The cavity effect narrowed Er^{3+} ion emission band $4I_{13}/_2 \rightarrow 4I_{15}/_2$ on intensity enhancement providing a Q factor of 171 with a cavity resonance at 1537 nm and stopband from 1350 nm to 1850 nm (MIR). Valligatla and co-authors [190] affirmed the importance of RF sputtering by fabricating a microcavity consisting of Er^{3+} doped SiO_2 active layer incorporated in between two Bragg reflectors now with ten pairs of SiO_2/TiO_2 layers as shown in Fig. 5. In contrast to [189], this small variation in the number of pairs resulted in the cavity resonance at 1749 nm and stopband present from 1500 nm to 2000 nm with a Q factor of 890 values. A dramatic reduction in the luminescence enhancement measurement error was also reported by the author for different detecting angle configurations.

A group of researchers [191–193] numerically modeled the Kerr nonlinearities in defective quadratic PC materials. The author theoretically studied that the enhanced optical limiting performance could be obtained by few parametric variations of PBG structure such as the number of periods, dimensions of PBG structure, and third-order optical nonlinearity of the composite material (see Fig. 7).

The revamping idea of reflective energy and power limiter was later introduced by Makri and his co-workers [182,194,195]. They presented the concept of incorporation of temperature-dependent loss tangent defect layer into a photonic crystal structure. This PC structure was having comparatively lower losses for an energy limiter in contrast to a power limiter where the nonlinearity of the defect layer is a prior necessity. The pulse duration also played a vital role in deciding whether the defect embedded layer structure would act as a power limiter (for long pulses) or energy limiters (for short pulses). Makri and co-authors proposed the limiting mechanism by resonant transmission phenomena through a localized mode. The nonlinear defect layer inserted in between two linear Bragg mirrors (lossless) as proposed earlier generated a localized mode with a central frequency lying within a photonic bandgap. The proposed architecture exhibited the resonant transmission at that resonant frequency with a field enhancement in the defect layer. Eventually, this enhancement resulted in an increase of the imaginary part of permittivity of the nonlinear defect layer, which further decoupled the Bragg reflectors, suppressed the localized mode and the limiter became reflective instead of absorptive or opaque. The advantageous feature of self-protection of this simple design also comes with the limitation of low light intensity transmittance into a narrow frequency range which can further be addressed by more sophisticated resonator crystal structures. Vella with his co-workers [195] implemented the architectural design proposed by Makri et al. [182,194] experimentally. Nonlinear defect layer of GaAs was embedded in between six consecutive alternate pairs of SiO_2/Si_3N_4 low loss linear materials constituting two Bragg reflectors, deposited by plasma-enhanced chemical vapor deposition (PECVD) technique. Transmission spectra of the designed structure depicted the photonic bandgap of Bragg grating between 1385 nm and 1970 nm with a central frequency of localized mode of the resonator at 1630 nm. The transmission coefficient got reduced from 0.41 (input peak power 4.24 GW/cm^2) to 0.026 (input power 46.3 GW/cm^2) at resonant transmission with a corresponding increase in the reflection coefficient from 0.24 to 0.98, indicating the proper functioning of the reflective optical limiter. Besides, a negligible damage threshold (visually inspected) also confirmed the uninterrupted optical limiting of the device from high irradiance.

The recent generation of novel protecting devices (optical limiters) is encouraging the photonic crystal structure for reflective optical limiters. Optical limiting in semiconductor nanocrystal incorporated in photonic bandgap structure [196], 3D polystyrene band-edge photonic crystals [197], 2D photo crystal structure in the visible region [198], and 1-D metal nanocomposites in photonic crystals [199] have been investigated by researchers. Liu and co-authors [200] theoretically analyzed the comparative optical limiting of a composite configuration constituted of a 1-D photonic bandgap and a thick bulk material by a backward split-step scheme. It was concluded that the limiting performance of compound configuration got improved by two orders of magnitude over the bulk material. It was also demonstrated that the increased number of periods (NOP) and the thickness of the defect layer had a significant effect on the efficiency of OL for further optimization of the composite network (see Table 3).

Ryzhov et al. [201] numerically stimulated and then experimentally

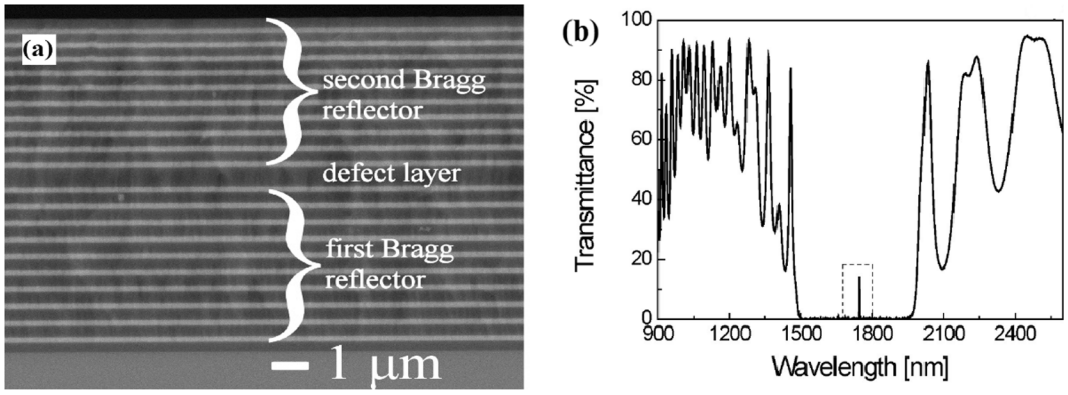

Fig. 7. (a) 1-D photonic crystal structures consisting of Er^{3+}doped SiO_2 defect layer and SiO_2/TiO_2 Bragg grating with substrate on the bottom (b) low-intensity transmission dispersion having peak at 1749 nm in the middle of photonic bandgap (1490–1980 nm). Reprinted with permission from [190], A. Chiasera et al. Opt. Express Vol. 20, (2012) © The Optical Society.

Table 3
1-D photonic crystal parameters.

Cavity Material	Bragg Reflector Material	Number of layers	Methodology	Cavity Resonance (nm)	Stop Band	Parameters	References
Er^{3+}doped SiO_2	SiO_2/TiO_2	25	RF Sputtering	1537	1350–1850	QF = 171 FWHM = 28 nm	[189]
Er^{3+}doped SiO_2	SiO_2/TiO_2	41	RF Sputtering	1749	1490–1980	QF = 890 FWHM = 1.97 nm	[190]
Nb_2O_5	Nb_2O_5/SiO_2	25	Ion Beam Sputtering	532.4	–	Tmax = 75%	[201]
ZnO	SiO_2/TiO_2	21	RF Sputtering	549	430–580	LT = 0.74 J/cm^2 Tmax = 39%	[203]
GaAs	SiO_2/Si_3N_4	25	PECVD/RF Sputtering	1633	1385–1970	FWHM = 13.3 nmTmax = 52%	[195]
GaAs	GaAs/AlAs	52	MBE	1116.8	–	LT = 1 mJ/cm^2 β = 25 cm/GW	[203]
GST ($Ge_2Sb_2Te_5$)	GST/SiO_2	9	RF Sputtering	1155	1250–2000	T = 84.7% in Pass state R = 89.3% in Block state	[204]

demonstrated the nonlinearities of Fabry Perot resonator consisting of 25 alternating layers of SiO_2 and Nb_2O_5 for 7 ns pulsed laser at 532 nm excitation wavelength. Nb_2O_5 being the middle thickest layer formed the cavity of the resonator along with other cavity layers by ion-assisted ion beam sputtering technique. Narrowing of transparency line with an increased reflection of resonant mirrors as a consequence of an increased number of layers was exhibited. Although fabrication of a good quality limiter with a transparency line below 1 nm of thickness emerges out as a technical challenge due to imprecise regulation of film thickness.

Semiconductors are well known for their inherent two-photon absorption nonlinear mechanisms and negligible linear absorption. This feature makes them competent for not only bulk optical limiters but also as a defect layer in a 1-D photonic crystal optical limiter [141,195]. Ryzhov [202] presented the GaAs semiconductor along with AlAs in the heterostructure microcavity in a more adequate manner for the protection of sophisticated optical devices. To date, the minimal limiting threshold reported was 0.01 J/cm^2 for phthalocyanines J-Type dimeric complex of Zn containing solution for 16 ns pulsed 532 nm lasers [83]. Ryzhov created a benchmark by reporting an exceptionally lower threshold value of 10^{-3} J/cm^2. Using the PECVM technique 52 layered microcavities consisting of GaAs/AlAs alternating layers and half-wave GaAs intracavity layer were fabricated. For nanosecond pulsed laser, TPA in addition to free career absorption contributed to refractive index variation of GaAs intracavity creating a resonance at 1.1 μm and reducing the transmittance to near about 4 times and hence altering the limiting threshold exceptionally low. Experimental evidence in favor of limiting properties of 1-D photonic crystal was evinced by Valligatla and co-workers [203]. A comparative study of a very popular ZnO semiconductor embedded photonic crystal and ZnO reference sample was performed. 21 alternating TiO_2/SiO_2 layered microcavity with ZnO half-wave defect layer inserted into the middle was fabricated by multi-target sputtering. TPA promoted the optical limiting in 1-D crystal structure exhibiting a threshold value as less than 0.74 J/cm^2 comparable to fullerenes, phthalocyanines, porphyrins kinds of nonlinear materials under 6 ns pulsed laser of 532 nm excitation wavelength and 10 Hz repetition rate. The transmission spectra displayed a wide working visible range with a stopband domain 430 nm to 580 nm and cavity resonance was centered at 532 nm, for a 30° incidence angle.

The resonant transmission-based 1-D photonic crystal optical limiter along with its benefits of self-protection also faces challenges of their angular sensitivity and narrow transmission width. In some applications like communication these limitations can be compromised but in spectroscopy and imaging kind of applications broadband transmission and angular sensitivity cannot be overlooked. Recently Sarangan and co-authors [204] exhibited a new approach of reflection-based optical limiter by using phase change material (PCM). Thin layers of GST ($Ge_2Sb_2Te_5$), chalcogenide glass capable of amorphous to crystal phase transition, were implemented along with alternate SiO_2 layers for fabricating 9 layered optical limiter using RF sputtering in an edge filter configuration. The authors demonstrated the switching of the device from PASS to BLOCK state with a temperature value rising above 145 °C. In PASS state the limiter exhibited the broad spectral bandwidth of 300 nm range (1500 nm-1800 nm) and large acceptance angle (±45°) and in BLOCK state represented the enhanced and omnidirectional reflectance illuminated by 1680 nm laser. The authors left few voids for researchers as BLOCK to PASS state conversion or resetting, transition dynamics, and damage threshold for future work.

The thick multilayered structure alters the switching speed of the

device and the nonvolatile phase changing of GST material compels the limiter to manually reset before switching it to the limiting state. Very recent C Wan et al. [205] demonstrated the ultrathin reflective optical limiting via resonant transmission achieved through metallic FSSs (frequency-selective surfaces). Insulator to metal transition (IMT) of vanadium oxide (VO_2) served as a huge nonlinearity source providing low Q factor FSS resonance. This ultrathin reflective limiter of thickness as low as 1/100 of the vacuum wavelength was fabricated by RF sputtering for 10.6 μm excitation wavelength. This designed limiter operated for polarization up to 50° incidence angle with a broader bandwidth (FWHM > 2 μm) having approximately 0.75 off-state transmittances, less than 0.01 on-state transmittance, and around 0.06 absorptance value. However, for high-intensity radiation, IMT in vanadium oxide could be triggered only tens of femtoseconds, limiting the device speed to microseconds only, left the scope of ultrafast vanadium oxide base reflective optical limiter design fabrication for the future.

5. Conclusion and future outlook

An extensive range of existing nonlinear materials has been boosted the enormous research in the field of optical limiters. An attempt has been made to review several nonlinear carbon structured materials such as fullerenes (0D), CNTs (1-D), graphene (2D), and carbon suspension nanoparticles (3D) along with semiconductors, chalcogenides, organometallics, nanodiamonds, nanorods, and many others. The limiting response of these materials extensively relies on assorted nonlinear processes. Third-order nonlinearity and SA in CNTs, RSA, and third-order nonlinearity in fullerenes and dyes, thermally induced NLS in CBS, CNT suspension, and in metal nanocomposites, MPA in semiconductors and chalcogenides, NLS and TPA in graphene has exhibited efficient optical limiting response with improved bandwidth and fast operating response. The problem of protecting the sophisticated optical device at the expense of limiting material loss has also been addressed and improved by recently studied 1-D photonic crystal (Reflection-based OL). To the best of our awareness currently invented reflection-based 1 D photonic crystal OL has yet not been reviewed. Our summary has emphasized effective outcomes of the 1-D photonic resonator for self-protecting optical limitation including phase-changing materials that have been exploited for the broad operating range of transmission. Numerous categories of nonlinear materials have exhibited their limiting threshold value remarkably below the injury threshold of the human eye or sensitive detectors, further efforts are required to set up their true preservative value. The broadband transmission and true protective threshold value are still a future goal to verify the realistic vulnerability of sensors. Further research is expected to examine the viability of phase change materials in real applications.

CRediT authorship contribution statement

Reena Gadhwal: Conceptualization, Methodology, Software, Data curation, Writing - original draft, Visualization, Investigation, Writing - review & editing. **Ambika Devi:** Investigation, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Reena Gadhwal received her B.Sc. degree from GNC Sirsa (KU University, Kurukshetra) India in 2007. She has completed her M.Sc. degree in Physics from Guru Jambheshwar University, Hisar, India in 2009. Now, she is a Ph.D. candidate in The NorthCap University, Gurugram, Haryana, India. Her research focus is in the field of Experimental Condensed Matter Physics and currently she is working on Optical Limiters.

Ambika Devi received her M.Sc. degree in Physics from Himachal Pradesh University, Shimla in 2006.She completed her PhD degree from JUIT, Waknaghat in April 2010. She joined The NorthCap University, Gurugram in July 2009. Presently, she is serving as Associate Professor in the Department of Applied Sciences. Her research interests include chalcogenide glasses, Polymer Nanocomposites and Supercapacitors.